

UC MATH 600 MATHEMATICS FOR COMPUTER SCIENCE · MODULE 12 · LECTURE NOTES

Expectation and Concentration

Averages, and why they are usually close to the truth.

MODULE OUTCOMES

- Compute expectations, including with linearity.
- Explain why linearity needs no independence.
- Use indicator variables to count.
- Apply Markov, Chebyshev, and Chernoff bounds.
- Explain concentration and why it matters.

1 Linearity

$E[X + Y] = E[X] + E[Y]$, ALWAYS — WITH NO INDEPENDENCE ASSUMPTION WHATSOEVER

It holds for dependent variables, for variables defined in terms of one another, for anything at all — which is genuinely remarkable and is used constantly without comment in every algorithms text you will read.

So a complicated random quantity can be broken into simple pieces, each of whose expectation is easy to compute, **and the pieces simply add — even when the pieces interact wildly with one another.**

Which is exactly what makes the indicator variable trick work (§2): write a count as a sum of zero-one variables and add up their probabilities, and you are done.

And note that $E[XY] = E[X] \cdot E[Y]$ does *not* hold in general — that one does require independence — and confusing the two is the standard error in this module, with the same shape as §4's variance asymmetry.

2 Indicator variables

THE TRICK

to find the expected NUMBER of things with some property, define

$X_i = 1$ if thing i has the property, else 0
 then $X = \text{sum of the } X_i$, and by linearity
 $E[X] = \text{sum of } E[X_i]$
 $= \text{sum of } P(\text{thing } i \text{ has the property})$

EXAMPLE: n letters into n random envelopes.

Expected number that land in the right envelope?

$X_i = 1$ if letter i is in its own envelope

$E[X_i] = P(\text{letter } i \text{ correct}) = 1/n$

$E[X] = n * (1/n) = 1$. Always exactly 1,
 for every n .

AND THE X_i ARE WILDLY DEPENDENT

if $n-1$ letters are correct then the last one
 must be too -- these are not independent in any
 sense -- and linearity does not care.

THAT INDIFFERENCE TO DEPENDENCE IS THE WHOLE POWER
 OF THE METHOD.

The indicators are wildly dependent and linearity does not care — which is what makes this the most reusable technique in the subject, since the dependence between parts of a problem is usually the hard part and this sidesteps it entirely. **Every hashing analysis in CSCE 629 and CSCE 658 is this trick:** expected collisions, expected bucket load, expected probe count — all of them are a sum of indicators.

3 Concentration

Bound	What it needs	What it gives
Markov	X non-negative, and its mean. Nothing else.	$P(X \geq a) \leq E[X]/a$. Weak, and free — and it is what the other two are proved from.
Chebyshev	The variance as well.	Deviating by k standard deviations has probability at most $1/k^2$ — which is distribution-free and useful.
Chernoff / Hoeffding	Independent, bounded random variables.	Exponentially small tails — vastly stronger, at the cost of the independence assumption.

Chernoff is exponentially stronger and needs independence — which is the trade, and is why the independence check of Module 11 §2 matters rather than being pedantry. **Markov from nothing, Chebyshev from the variance, Chernoff from independence:** three bounds, three price points, and knowing which you can afford is the practical skill.

AND CONCENTRATION IS WHY AVERAGES ARE USEFUL AT ALL

A sum of many independent small contributions lies very close to its mean with overwhelming probability — which is why sampling works, why polling works, and why randomised algorithms are reliable despite being random.

And the probability of being far from the mean falls exponentially in the number of trials — so a few hundred samples pin down a proportion remarkably well, which is why survey sample sizes are smaller than intuition suggests.

Which is the formal content of the vague claim that 'the average is representative' — a statement that is simply false for a single trial and true for many, and the bounds say precisely how many you need for a given confidence.

And it is what CSCE 658 runs on: repeat a randomised algorithm k times and take the majority, and the failure probability falls like a constant raised to the k — which converts an unreliable procedure into a reliable one at logarithmic cost, and is the central move of that whole course.

4 Variance

- **$\text{Var}(X) = E[X^2] - E[X]^2$, which is the computational form and is how it is always actually calculated, the definition as $E[(X - \mu)^2]$ being harder to work with.**
- **And $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ only when X and Y are independent — unlike expectation (§1) — which is the asymmetry to remember and the one people get backwards.**
- **In general $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$, and the covariance term is exactly what correlated failures contribute** (Module 11 §4's independence error) — positive covariance means the deviations reinforce.

- Which is why diversification works and why correlated risk defeats it — the same mathematical fact, appearing in finance and in system reliability, and in both places the covariance is the term nobody measured.
- And the standard deviation of a sum of n independent terms scales as $\sqrt{n}$, so the *average's* standard deviation shrinks as $1/\sqrt{n}$ — which is the square-root law behind every sample size calculation: quadrupling your sample halves your error, which is why precision is expensive.

AND THIS IS THE END OF THE PREREQUISITE

Linearity of expectation, the indicator trick, and the three concentration bounds are the whole probabilistic toolkit that CSCE 629 and CSCE 658 assume — and they assume it without comment, which is the attrition mechanism Module 01 §4 described.

The expected running time of quicksort is an indicator argument; the maximum load of a hash table is a Chernoff bound; amplification by repetition is concentration — three sentences from the first half of Semester 1.

So if those three sentences make sense to you, this course has done its job — and if they do not, Modules 11 and 12 are worth another week before starting, which is a cheap insurance premium.

Which is Module 13's self-check, essentially — and taking it honestly is the last thing this prerequisite asks of you.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 19 and 20 (free)	The whole module , free — and the indicator variable examples are the best collection available.
Mitzenmacher & Upfal, chapters 2 to 4	§§1 to 4 — expectation, variance, and the Chernoff bounds proved. The direct bridge to CSCE 658. Library copy.
MIT 6.042J, the expectation and deviation lectures (free)	§§1 to 3, with linearity's independence-freedom emphasised properly.
Motwani & Raghavan — Randomized Algorithms	§3 applied — the bounds used in anger, which is CSCE 658's content. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Compute five expectations** directly from the definition.
- 2 **Prove linearity** for two dependent variables.
- 3 **Find a case where $E[XY] \neq E[X]E[Y]$.**
- 4 **Solve the envelope problem** by indicators.
- 5 **Count expected collisions** in a hash table by indicators.
- 6 **Apply Markov, Chebyshev, and Chernoff** to the same problem and compare.
- 7 **Find how many samples** you need for 1% precision at 95% confidence.
- 8 **Compute a variance** both ways and confirm they agree.
- 9 **Find two correlated variables** and compute the covariance term.
- 10 **Verify the $1/\sqrt{n}$ law** empirically by simulation.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State linearity of expectation and what it does not assume.
- 2 Which product rule does need independence?
- 3 Describe the indicator trick and why linearity enables it.
- 4 Solve the envelope problem and say why dependence is irrelevant.
- 5 Name the three bounds and what each requires.
- 6 Why is Chernoff worth its assumption?
- 7 What is concentration, and why does it make averages useful?
- 8 Give the variance computational form.
- 9 When does variance add, and what is the extra term otherwise?
- 10 State the square-root law and its practical consequence.